

Comparing game enjoyment, competitive anxiety and performance of sit-up exercise via a team competition app

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ABSTRACT

Exergames are used to achieve some types of outcomes in addition to offering entertainment. To examine the practice effect of exergaming, this study had 6th grade students compete collaboratively in a sit-up competition using the Fit-Fun exergame app. Each participant performed sit-ups for 1 min, and five participants' sit-ups were accumulated to calculate the group's score. Applying achievement emotion theory, this study examined the change in the students' game enjoyment, competition anxiety, and exercise performance over six trials. A single-group quasi-experimental design was conducted, and students were encouraged to take part via the purposive sampling method. Questionnaires were completed at the end of each of the six trials. A total of 208 valid samples were collected for statistical analysis. Results showed a noteworthy improvement in exergaming performance, but a decrease in game enjoyment, while game competition anxiety stayed relatively stable. The findings contribute to the literature by applying achievement emotion theory to a collaborative exergaming context, and showing how constant competitive gameplay can increase performance while potentially weakening enjoyment. The findings may be a future exergame and physical education design reference.

1. Introduction

"Exergames" are those games that involve exercise while playing; they are also known as gaming for health. Exergame interventions emphasize ease of use and accessibility through a human-centered design, while offering progressively increasing challenges [1]. The mechanics of the game, or the game play method, require movement of the body while playing, thus increasing physical activity levels. For example, exergaming may change users' health-related behavior (e.g., [2,3]), and is also known to create extensive psychological effects [4]. Through enactive realism in exergames, people receive direct access to the mind-independent physical world as they engage in exercise [5].

Combining the Internet of Things (IoT) with gamification serves to promote the creation of entertaining and accessible exercise tools that suit a majority of individuals, irrespective of their physical or cognitive capabilities; for example, such tools can be used for remote rehabilitation (Mulholland & Merilampi [6]). Although previous studies have shown that exergaming can improve physical activity and motivation (e.g., [7,8]), few have unified a theoretical framework including achievement emotion theory [9] to explain the emotional changes involved in

repeated competitive gameplay. There is therefore a gap in understanding of the emotional changes that take place when students use the IoT to do exercise.

Exergames that combine physical activity with gaming offer an alternative avenue for exploring players' emotional changes [10]. According to achievement emotion theory, emotions are either positively activating (e.g., enjoyment) or negatively deactivating (e.g., anxiety) [11]. Hassan et al. [12] studied the outcomes of exergaming in fitness training, and found that exergaming not only enhanced students' energy, but also improved their emotional level compared to traditional physical exercise programs. Moreover, technology has progressively been utilized to support physical activity and regulate emotion [13,14]. For example, previous studies have investigated how intelligent systems and IoT applications can monitor and manage depression, anxiety and stress in real-time exercise [15].

However, few studies have focused on designing exergaming with the IoT to explore the emotional factors in collaborative exercise competitions. Thus, this study addressed this gap by applying achievement emotion theory to evaluate changes in students' performance and in their emotional outcomes over a series of trials. That is, the present

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study applied an exergaming device, named Fit-Fun. When playing, players wear a belt with a pressure sensor while performing sit-ups, and a rabbit avatar in the app advances one step for each sit-up.

1.1. The Purpose of this study

According to achievement emotion theory (AET; [9]), negative emotions include boredom, anxiety, and anger, while positive emotions include enjoyment and gratitude. Emotions such as performance anxiety and anger [16,17,18] may provide insights into understanding the competitive and cooperative frameworks characteristic of educational games [19].

Both positive and negative emotions can be facilitated by game-based learning environments. For example, when players cooperate while playing a game, they may face negative experiences with their cooperative peers or teammates (e.g., fault attribution, conflict and frustration) [20] as they form squads or teams to work together to accomplish gaming goals [21]. In terms of emotions in sports, athletes also experience them during competition, some of which may be pleasant (e.g., enjoyment and happiness), while at times others may be unpleasant (e.g., anxiety or anger) [22]. Previous systematic reviews determined that psychological emotions are influenced by exergaming [23].

Cooperation while competing is a basic mode of social interaction that generates diverse emotional and cognitive experiences [24]. The experimental design in this study used AET to explore game features (e.g., sit-up frequency translated into the jumping of an avatar rabbit), schemes of competitiveness, and emotion in the gameplay experience to explain performance outcomes. According to AET, the perspective of “multiple emotions” states that emotions interact with each other and co-occur within a situation or within a person [25]. Research assuming this viewpoint pays attention to finding patterns or profiles of emotions changing within a situation or person to examine the change in emotion patterns [26]. That is, rather than experiencing a single distinct emotion, people can experience numerous emotions concurrently in the same context. Thus, the aim of this study was to reveal the change in players’ positive and negative emotions, and in their achievement as they used the Fit-Fun app.

Most digital games that pit people against other players or machines revolve around competition [27]. Accordingly, the present study can provide a comprehensive perspective on the emotions experienced while involved in an exergaming collaborative competition. Moreover, the goal of this study was to provide valuable insights for future understanding of emotion change through the lens of AET in order to motivate students to engage in exergaming.

1.2. Research questions

1.2.1. Game enjoyment

When designing games according to the requirements and recommendations of player enjoyment evaluation models, educational game developers and designers can efficiently shape their games to prioritize the specific aspects of enjoyment they aim to foster in gamers [28]. This allows them to make informed decisions about the design of the games, thereby improving the users’ overall gaming experience De Almeida & dos Santos Machado [29]. Exergaming used to be centered on participants’ enjoyment of videogames and its enjoyable and engaging features. For instance, utilizing a design-based exergame as a strategy can be effective in terms of encouraging students to engage in physical exercise [30,31]. Indeed, emotions are brought about by a specific situation; in some competitions, emotions can be triggered more strongly, but then become weaker in intensity, while in some cases, they may persist for longer [32].

Gaming in general can reduce stress and improve hedonic and eudaimonic well-being (e.g., [33]). Gamers’ evaluation models of enjoyment allow developers and designers to understand how game

performance promotes gamers’ enjoyment De Almeida & dos Santos Machado [29]. It is possible that the perception of game realism makes it easier to match with others in a competition between the game and the other team’s performance, thus allowing the players to focus more on the game’s challenges [34]. The goal to compete with other teams can encourage players’ engagement and enhance their enjoyment of the gaming experience. Even in temporary teams when the goals of a specific challenge are achieved, such game-based team competition affects the enjoyment of everyone in the online gaming community [35].

Current research on emotions has demonstrated that emotional experience is rather more fluid, comprising a dynamic relationship of assessments and evaluation, with emotion as a construct dimension that can change over circumstances and time [36]. Moreover, emotions such as “anxiety” or “joy” are not distinctive classifications, but are somewhat changing and defining forms of emotional experiences happening at a given time [37]. Therefore, this study explored how emotions changed as the competition time increased. Accordingly, the Fit-Fun exergame app was used in this study as a useful training aid to improve students’ sit-up performance. A research question was proposed to understand to what extent their game enjoyment increased as the number of competition times increased while they were playing the exergame in team cooperating mode and competing against other teams using electronic devices.

RQ1: Does players’ game enjoyment change as the number of competition times increases?

1.2.2. Competitive anxiety

Most studies on competitive anxiety have primarily focused on state anxiety, which represents a more situation-specific and temporary form of anxiety [38]. Hong et al. [39] discussed the value of competition in games, stating that students engage in play activities when a competition is intense. A previous study revealed that high anxiety circumstances cause athletes to lessen their timing performance expectations [40]. For example, individuals who participate in high-pressure contests such as cognitive games involving time limits or competitive interactions tend to create a heightened interest in strategic decision-making. However, when it comes to competing for rewards, the competitive nature of such contests often causes emotional anxiety among participants (Shi et al. [41][42]). To investigate the developing emotional patterns that naturally emerge during competitive sessions, studies have identified distinct subgroups of athletes who heavily rely on competition [43]. The emotion could be increasing (incline shape through time), decreasing (weakening shape through time) or stable (no change through time) as it positively or negatively changes across time [44].

Previous studies have examined numerous approaches employing different devices to assist athletes in reducing the intensity of their competitive anxiety or advancing more constructive understandings of anxiety symptoms [45]; Mellalieu et al. [38]. Accordingly, a research question was proposed to understand the fluctuation of gameplay anxiety in the team playing of Fit-Fun.

RQ2: Does players’ gameplay anxiety change as the number of competition times increases?

1.2.3. Exergaming performance

Exergaming performance usually relies on the expertise and criteria of the professionals who manage the training program, rather than being guided by objective standards and validated performance measurements [46]. Exergaming performance refers to the inclination of a participant to use their physical strength, accuracy, rhythm, balance, flexibility and endurance to try to attain positive outcomes during exercise competition [47]. Reynolds et al. [48] indicated that those who regularly exercise in a physical setting gain from exergaming situations due to the degree of variance found in the routine of movement patterns and physical activity. However, competing against others leads to better performances, but it can also demotivate and reduce players’ performances in

exergames [49,50].

Furthermore, an increase in the practice duration can significantly enhance performance and play a critical role in numerous aspects of the resulting learning curve, including anxiety levels. How many practices bring out the best performance? One perspective recommends that a power function delivers the best fit, indicating that performance improvement is most significant in the early stages of practice, and then gradually plateaus as practice time increases [51]. This perspective is regarded as the “power law of practice” [52,53]. This study focused on the team challenge of sit-up frequency (i.e., how many sit-ups students in a team of five can perform) while in the team competition. As the accuracy of sit-ups is considered a challenge in exergame interventions, for each accurate sit-up a team performed, the rabbit was triggered to jump a step forward in the race, as shown on the app interface. Thus, a research question was proposed to explore the exergaming performance in this study, specifically the number of accurate sit-ups performed in one minute.

RQ3: Is there a power law of practice for participants as their exergaming times increase?

2. The current study

The current study began recruited sixth-grade students, who were introduced to the Fit-Fun exergaming App, and they then participated in a team-based sit-up competition across six sessions. After the session, they were asked to complete questionnaires that measure game enjoyment and competition anxiety, while their performance was recorded through the App as summarized in Fig. 1.

2.1. The fit-fun app

The physical fitness app, Fit-Fun, created by the Digital Learning Experiment Research Team, allows users to perform sit-ups using an exercise belt. In Figs. 2 and 3, the testing environment is displayed. The player ties the workout belt around their waist during play, 3 to 5 mm above the hip bone so that the sensor gadget is positioned 1 to 2 cm below the navel. There are single player and multiplayer modes in the Fit-Fun sit-up game. Six weeks’ worth of teaching sessions totaling six trials of gameplay were carried out in this study.

The first player in the “5 Player Battle” option enters the gaming environment, chooses the avatar (a rabbit), and then chooses the minute competition period. The first participant must access the challenger room for the game and wait for opponents to join. The tablet will begin a count down from 3 s when the first participant pushes the begin button, giving everyone else 3 s to prepare. Each player must do sit-ups until the completion of the countdown minute, or until every member of the team has participated. The rabbit moves forward one step after each sit-up.

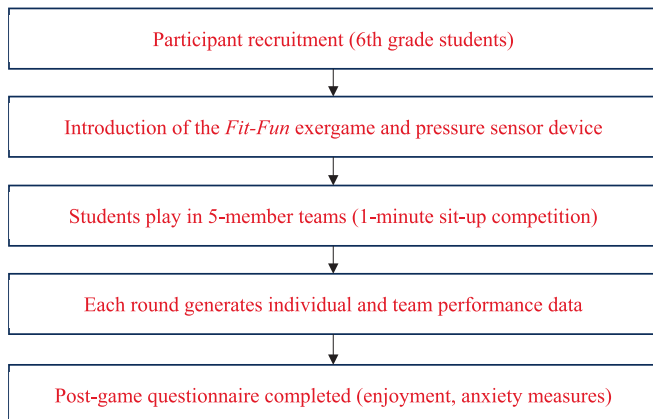


Fig. 1. Overview of the study procedure.

The game will stop instantly after the allotted time has passed, and the outcome, the total number of sit-ups performed along with the placing, will be displayed on the tablet as the competition’s benchmark results.

2.2. Procedure and participants

The participants in this study were elementary school sixth-graders. A single-group quasi-experimental study took place from February to March 2023, and participants were chosen from eight classes using the purposive sampling method. First, to motivate students to compete collaboratively, five students were randomly grouped into teams; second, participants took turns doing sit-ups for 1 min while wearing the belt. Third, the frequencies of each player’s sit-ups were recorded and shown on the iPad, and the results were summed to represent the team’s performance to decide the winning team. Fourth, at the end of each trial, questionnaires related to gameplay enjoyment and anxiety were collected from participants.

Considering ethical issues, each parent of the participant provided their informed consent. This study was approved by the Ethical Committee with the REC number 201812HS020. Students were aware that they were taking part in an evaluation study and that the information they submitted would be kept anonymous.

Students from the eight classes were required to take part in the team sit-up competition, except for those younger children with immature large muscle development. In all, 208 valid samples were collected, of whom 98 (46.9 %) were male and 110 (53.1 %) were female. They were 11 years old (+ 0.4).

2.3. Questionnaire

2.3.1. Development of the questionnaire

The questionnaire for this study was created with reference to pertinent literature regarding game satisfaction and rivalry apprehension. First, we recruited three subject-matter experts to confirm the precision of the item translation and the items’ relevance. Second, to check the questionnaire’s face validity, 10 students were asked to pretest it. The degree of agreement was assessed using a 5-point Likert scale, where 1 represented *strongly disagree* and 5 represented *strongly agree*. In addition, the outcomes of the tournament included exergaming performance. The reliability of the items and the constructs was tested after the face validity examination by the experts.

2.3.2. Measurement

Game enjoyment: Adapting from the study of Reer and Kramer [54], we measured game enjoyment using six items which included statements such as “I really enjoyed playing the exergame” and “I think I am happy when playing the sit-up game.” Internal reliability Cronbach’s $\alpha = 0.90$ and composite reliability (CR) = 0.91.

Competition anxiety: A competitive anxiety measurement tool was developed by modifying the anxiety questionnaire created by Hong et al. [39]. Example items are “I would be too worried about losing the competition during the game” and “I get flustered if I cannot do sit-ups faster than others”; Cronbach’s $\alpha = 0.92$ and CR = 0.92.

2.3.3. Data analysis

This study adopted a within-subjects quasi-experimental design to evaluate the effects of repeated competitive exergaming. Participants participated in the exergame over six sessions, and data were collected immediately after each play using a structured questionnaire measuring participants’ game enjoyment and competition anxiety. Objective performance data (sit-up counts) were recorded directly from the Fit-Fun gaming system. The evaluation included statistical comparison of changes across the six sessions using paired sample *t* tests and repeated measures ANOVA.



Fig. 2. Fit-fun and exercise belt.

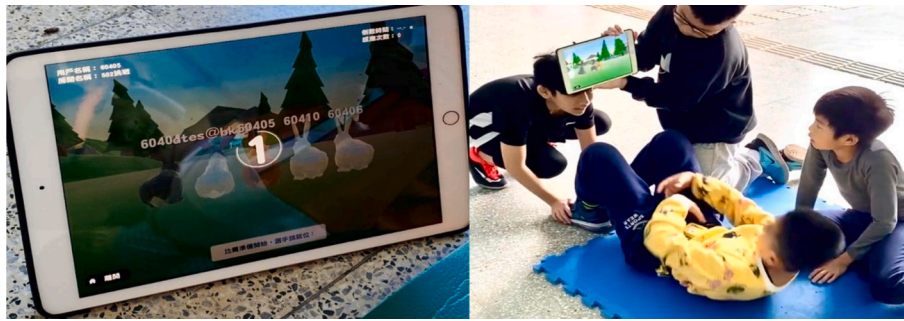


Fig. 3. Participant experience and gaming environment.

3. Results

3.1. Exergaming progress analysis

To identify the performance of each trial, this study recorded the frequency of every participant's sit-up performance. The analysis of the paired sample t test revealed that the first trial's accuracy rate was $M = 21.12$, $SD = 10.36$, and the fifth trial's accuracy rate was $M = 35.02$, $SD = 13.03$, with $t_{(67)} = -9.44^{***}$, $p < 0.001$, which was significant. In the first trial, the medium value was 17.6 and the mode value was 18.0. In the sixth trial, the medium value rose to 34.09 and the mode value was 26.0. Table 1 presents the mean (M), standard deviation (SD), plural (Mode), and median (see) of the sit-up frequency in one minute in the first and sixth trials.

3.2. Emotion and performance changes with practice time

To understand the participants' emotional changes after experiencing the Fit-Fun sit-up challenge, a time-series statistical analysis was conducted on their game situational interest, game competition anxiety, and exercise performance. Changes in the cognitive and affective factors were compared by means of time series analysis, and sit-up performance was divided by five for each of the average scores obtained.

The blue line represents game situational interest; in the pretest, $M = 4.20$, $SD = 0.80$, while in the posttest it decreased to $M = 3.79$, $SD = 0.93$, with a t -value of 3.72 and a statistical significance of $p < 0.001$, $d = 0.47$, representing a medium-high effect. The red line represents game competition anxiety; in the pretest, $M = 2.51$, $SD = 1.22$, while in

the posttest it decreased to $M = 2.32$, $SD = 1.39$, with a t -value of 0.67, which was not statistically significant. The yellow line represents exercise performance; in the pretest, $M = 21.12$, $SD = 10.36$, Mode = 18, Medium = 17.76, while in the posttest it increased to $M = 35.02$, $SD = 13.03$, Mode = 26, Medium = 34.09, with a t -value of 4.94 and a statistical significance of $p < 0.001$, $d = 1.35$, which represents a high effect (see Fig. 4 and Table 2). Moreover, Table 3 shows the relationship between research questions, measured variables, and findings.

4. Discussion

According to achievement emotion theory (AET; [9]), emotions are divided into positive and negative emotions, which may provide insights into understanding the competitive and cooperative frameworks characteristic of educational games [19]. Drawing on AET, this study focused on the change in students' game enjoyment, competitive anxiety, and exergaming performance while they performed sit-ups in teams wearing pressure sensors to detect their sit-up frequency. The three research questions are answered as follows.

4.1. Game enjoyment analysis

Gaming has been found to reduce individuals' stress and to contribute to hedonic and eudaimonic well-being [33]. Evaluation models of player enjoyment have been developed to measure the performance of games for promoting player enjoyment and finding areas for improvement De Almeida & dos Santos Machado [29]. Perceived game realism can increase competition matching and allow players to concentrate more on the game's challenges, which leads to increased engagement and enjoyment [35]. Game designers and developers can apply player enjoyment evaluation models to guide their decision-making process and highlight the characteristics of enjoyment they intend to promote in educational games De Almeida & dos Santos Machado [29]. Exergames combine exercise with gaming, and have been used to support students' interest in physical activity [30].

Table 1

Comparative analysis of sit-up performance.

Trial number	M	SD	Medium	Mode	t	d	Result
First	21.12	10.36	17.76	18.0	-9.44***	1.35	$2 > 1$
Sixth	35.02	13.03	34.09	26.0			

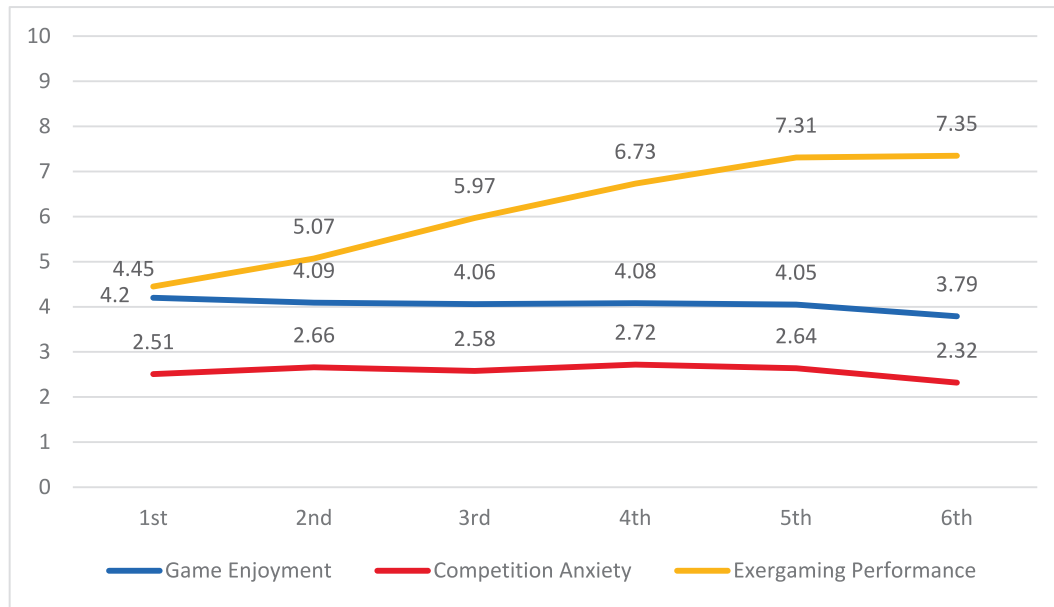


Fig. 4. Time series analysis of game enjoyment, competition anxiety, and exergaming performance.

Table 2

Comparative analysis of game enjoyment, competition anxiety, and exercise performance.

	Trial	<i>M</i>	<i>SD</i>	<i>t</i> -value	<i>d</i>	Comparison
Game enjoyment	First	4.20	0.80	3.72***	0.47	1 > 2
	Sixth	3.79	0.93			
Competition anxiety	First	2.51	1.22	0.67	–	–
	Sixth	2.33	1.39			
Exercise performance	First	21.12	10.36	4.94***	1.35	2 > 1
	Sixth	35.02	13.03			

Table 3

Relationship between research questions, measured variables, and findings.

Research Question (RQ)	Variable measured	Key finding
RQ1: Does game enjoyment change with repeated competition?	Game Enjoyment (Self-report)	Enjoyment decreased significantly over time
RQ2: Does gameplay anxiety change with repeated competition?	Competitive Anxiety (Self-report)	Anxiety remained relatively stable
RQ3: Is there a power law of practice for performance improvement?	Sit-up Performance (App Data)	Performance significantly improved over time

Regarding game enjoyment, the participants reported $M = 4.20$ with a $SD = 0.80$ during the first trial, indicating a high level of enjoyment. However, in the sixth trial, the mean score $M = 3.79$ decreased slightly with $SD = 0.93$. The t -value of 3.72 suggests a significant difference between the first and sixth trials, indicating a decline in game enjoyment over time. The decrease in game enjoyment may be caused by numerous factors, such as repeating the gameplay process, knowledge of the challenges, or other external factors that were not analyzed in this study. Thus, it is essential to realize how game enjoyment changes with increasing competition time.

4.2. Competitive anxiety analysis

Competitive anxiety, frequently studied in the setting of state anxiety, is often regarded as situation-specific and is known to be a short-term form of anxiety [38]. Competitive circumstances can provoke

engagement in playing activities and can affect performance outcomes Hong et al. [39]. Athletes' anticipated timing performance can be affected by high anxiety levels [40]. However, competition can motivate interest in strategic decision making, and may also stimulate emotional interest among participants, specifically when the competition includes rewards (Shi et al., 2021). A previous study showed that there are different emotional trajectories across competitive events, with emotions displaying accumulative, lessening, or stable patterns over a period of time [44]. One study previously showed that using different devices can decrease competitive anxiety or support more facilitative understandings of anxiety [38]. In terms of competitive anxiety, the participants reported similar levels of anxiety during the first ($M = 2.51$, $SD = 1.22$) and sixth ($M = 2.33$, $SD = 1.39$) trials of exergaming.

4.3. Exergaming performance analysis

Exergaming performance is affected by numerous factors such as strength, power, endurance, flexibility, balance, rhythm, and precision on the physical plane [47]. Exercising or practicing physical activities in real-world environments has been revealed to benefit performance in exergaming environments [48]. However, challenging or competing with one another can both demotivate and motivate players' performance in exergames [49,50]. Practice can lead to significant improvements in performance. However, there is a learning curve where performance gains begin to level off over time [51]. According to previous studies, performance improves the most during the early stage of practice and will progressively plateau as practice time increases [52]. The participants' exercise performance was measured according to the number of sit-ups completed, and showed a significant improvement from the first to the sixth trial. The mean score $M = 21.12$ increased with $SD = 10.36$ during the first trial to $M = 35.02$ with $SD = 13.03$ during the sixth trial. The t -value of 4.94 showed a significant difference in exergaming performance between these two trials, signifying that participants' physical capabilities improved over the course of the study. This improvement may be because of the participants' increased knowledge of the Fit-Fun system, as well as their motivation to compete against other teams, which led to an improvement in their performance. This study focused on a sit-up frequency team competition using the Fit-Fun app, with the accuracy of sit-ups as the performance indicator.

In summary, this study provides several contributions to the literature. First, repeated play of competitive exergaming can significantly

improve students' physical performance, suggesting it has great value as a tool for improving students' physical fitness in school settings. Second, the decline in game enjoyment over time points out a potential disadvantage of competitive gaming, highlighting the need to cautiously manage challenge levels and novelty in game design. Third, the stability of competition anxiety specifies that students adapted to the competitive environment without experiencing sensitive stress, which supports the capability of short-term repeated competitions in physical education. Overall, these findings underline the importance of balancing the outcome of performance with emotional engagement to confirm sustainable enjoyment and motivation in game-based exercise programs.

5. Conclusions

Collaboration allows individuals to work together to achieve better goals. Collaboration among participants allows for the achievement of common goals, leading to the performance of each participant being improved due to others' encouragement. Group competition always has the common feature of social interaction. In other words, the Fit-Fun app allowed players to compete as a team of five, and the group's total sit-up count was used to determine the winning team. This collective goal may have fostered group cohesion, social accountability and peer encouragement, all of which can motivate students to increase their efforts during gameplay. The sit-up exergame was designed such that some degree of performing more sit-ups is required to achieve the team objectives and emotion change. This study showed that the perceived realism of a game makes it easier to match up with others in a competition, and enables players to concentrate on the challenges of the game. Additionally, this study showed that in team-based competitions in online gaming communities, there is a high level of enjoyment among players when team goals are achieved, but this enjoyment slightly declines as the number of practice times increases. On the other hand, in this study, the participants' gameplay anxiety only slightly decreased in the collaborative competition.

The participants' exergaming performance was affected by several factors, including their experience of the exergame, practice, peer motivation, the competition effect, and immediate feedback from the Fit-Fun app. Participants' intention to strive for specific goals, such as physical strength, speed, power, endurance, flexibility, and so on are some criteria that contribute to their performance in exergames. Moreover, competing against one another can enhance performance. However, such competition can also demotivate and reduce participants' overall performance. Practicing extensively plays an important role in improving exergaming performance, with the power law of practice suggesting that performance improves the most in the early stages of practice, and begins to level off as practice time increases.

Exercises that are enhanced by technology and that encourage physical activity are becoming increasingly popular, and seem to foster some people's willingness to continue practicing. For instance, mobile phone apps designed to improve exercise participation are frequently built on an achievement system such as earning points for completing a fitness goal. The Fit-Fun sit-up physical fitness app and exercise belt used in this study proved to be a meaningful tool for improving students' sit-up performance.

5.1. Theoretical and practical implications

The implications of this study are twofold. First, this study extends our knowledge of students' achievement emotion and performance by examining their gameplay enjoyment and anxiety, and their practice progress. Measuring the change in achievement emotion is critical in exergaming research because different gamified features may closely affect performance. Scholars have focused more on the cognitive and affective dimensions at the final stage or trial, but less on the change dimension with increase in the number of exercise times. However, this study focused on the possibility that exergaming might be affected by

experiences with gamified features. In Fit-Fun, in which students perform sit-ups which trigger an avatar rabbit to jump, the change in achievement emotion was examined one trial at a time, and it was found that gameplay enjoyment and anxiety slightly decreased as the number of practice times increased.

Furthermore, the results of this study support the power law of practice when conducting exergaming interventions. It is acknowledged that performance improvement may be most significant during the first phase of practice, and then tends to level off with increased practice time. When exercising with peers, emotion may be especially responsive to peer input during collaborative competitions. The social aspect is important in many forms of exercise, and the results of this study support this perspective, as the participants clearly preferred to perform sit-ups with partners in the formal collaborative competition setting.

Second, this study expands the understanding of exergaming performance, particularly focusing on students collaboratively performing accurate sit-ups in one minute. However, some students may prefer to exercise alone before joining a collaborative competition if they believe they cannot perform sit-ups as well as other students. Thus, we suggest giving students opportunities to exercise alone before performing in group settings. Thus, achievement emotion in a social setting can be compared to the pre-ability of performing sit-ups via the Fit-Fun app.

5.2. Limitations and future study

The findings may be limited to the specific context of the Fit-Fun exergaming system and the participants involved. The sample size and features of the participants might not characterize the wider population of gamers or users who engage in other exergaming activities. Therefore, caution should be taken when generalizing the results to other settings or populations. Future studies can investigate the effects of the Fit-Fun exergaming app on a different population range, such as participants with different fitness levels, and of different age groups and cultural backgrounds. In this way, we can understand potential differences in game enjoyment, competitive anxiety, and exergaming performance across diverse populations.

Extensive practice significantly improves skilled performance, with a key focus in numerous fields being the shape of the resulting performance curve. This perspective aligns with the "power law of practice" [52]. This study was conducted over a relatively short period of six teaching sessions conducted over 6 weeks. This short period might not be sufficient to help us understand the long-term effects or changes in participants' game enjoyment, competitive anxiety, and exergaming performance. Future studies may consider longer follow-up periods such as longitudinal studies with extended follow-up periods which can offer insights into the long-term effects of users' exergaming on game enjoyment, competitive anxiety, and performance.

Future studies may also use mixed methods approaches to conduct quantitative measures and qualitative data collection methods, such as interviews or focus groups which can deliver a more wide-ranging understanding of participants' experiences and perceptions. This can also help reveal nuanced aspects affecting game enjoyment, competitive anxiety, and exergaming performance based on AET. In addition, the current study relied on self-reported measures to assess emotional states; future studies may benefit from integrating technologies such as facial expression analysis or physiological monitoring which may offer opportunities to more accurately and continuously measure emotional responses during gameplay [55].

Informed consent.

Considering ethical issues, each participant and their parents provided their informed consent. The Ethical Committee of National Taiwan Normal University with REC number 201812HS020 approved this study, indicating that students knew that they were participating in an evaluation study and that the data they provided would remain anonymous.

CRediT authorship contribution statement

Wei-Chen Hung: Writing – original draft, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Jon-Chao Hong:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Funding acquisition, Conceptualization. **Yu-Feng Wu:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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